PDF Viewer Test 12: zoom stem-symmetry diagnostic

PAGE 1/3 -- CONTROL: no /Widths array, fontWidthFactor forced to 1.0 (no stretch).

View this document at 100% zoom first, then switch to 125% zoom and compare the SAME page. Look specifically at the vertical stems of H, N, M, I, l, T, W for left/right thickness asymmetry.

Hypothesis under test: main/gfx.goc's setTextTransform() computes font->fontWidthFactor ONCE per font, from the ratio of the PDF's declared width for 'm' vs. the GEOS substitute font's actual rendered 'm' width, then bakes it in as an INTEGER percentage via GrSetFontWidth(gstring, IntegerOf(font->fontWidthFactor * 100)). Because pages are cached (Phase 3), zooming later replays the same GString without recomputing this factor -- so the prediction is: pages with fontWidthFactor far from 1.0 should show WORSE stem asymmetry at 125% than the control page, which should look clean at both zoom levels.

HNMI lTW

HNMI lTW

HNMI lTW

HNMI lTW

HNMI lTW

HNMI lTW

IIIIIIII llllllll MMMM HHHH

PDF Viewer Test 12: zoom stem-symmetry diagnostic

PAGE 2/3 -- ’m’ declared at 500/1000 em (real Helvetica ~889) -> ~56% width factor.

View this document at 100% zoom first, then switch to 125% zoom and compare the SAME page. Look specifically at the vertical stems of H, N, M, I, l, T, W for left/right thickness asymmetry.

Hypothesis under test: main/gfx.goc’s setTextTransform() computes font->fontWidthFactor ONCE per font, from the ratio of the PDF’s declared width for ’m’ vs. the GEOS substitute font’s actual rendered ’m’ width, then bakes it in as an INTEGER percentage via GrSetFontWidth(gstring, IntegerOf(font->fontWidthFactor * 100)). Because pages are cached (Phase 3), zooming later replays the same GString without recomputing this factor -- so the prediction is: pages with fontWidthFactor far from 1.0 should show WORSE stem asymmetry at 125% than the control page, which should look clean at both zoom levels.

HNM I l TW

HNM I l TW

HNM I l TW

HNM I l TW

HNM I l TW

HNM I l TW

I I I I I I I I l l l l l l l l MMMM HHHH

PDF Viewer Test 12: zoom stem-symmetry diagnostic

PAGE 3/3 -- ’m’ declared at 300/1000 em -> ~34% width factor (near the compression floor).

View this document at 100% zoom first, then switch to 125% zoom and compare the SAME page. Look specifically at the vertical stems of H, N, M, I, l, T, W for left/right thickness asymmetry.

Hypothesis under test: main/gfx.goc’s setTextTransform() computes font->fontWidthFactor ONCE per font, from the ratio of the PDF’s declared width for ’m’ vs. the GEOS substitute font’s actual rendered ’m’ width, then bakes it in as an INTEGER percentage via GrSetFontWidth(gstring, IntegerOf(font->fontWidthFactor * 100)). Because pages are cached (Phase 3), zooming later replays the same GString without recomputing this factor -- so the prediction is: pages with fontWidthFactor far from 1.0 should show WORSE stem asymmetry at 125% than the control page, which should look clean at both zoom levels.

HNM I l TW

HNM I l TW

HNM I l TW

HNM I l TW

HNM I l TW

HNM I l TW

I I I I I I I I l l l l l l l l MMMM HHHH